

# Absorption Signals

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// Absorption Signals

// Version: 1.0

// Author: QuantShok (JacobS369)

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// (in whole or in part), you must credit the original author:

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// Original script by QuantShok (JacobS369)

// <https://www.tradingview.com/u/JacobS369/>

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// Default settings are optimized for NQ (Nasdaq 100 Futures).

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//@version=6

indicator("Absorption Signals", shorttitle="Absorption", overlay=true, max\_labels\_count=500)

import TradingView/ta/8

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// INPUTS

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var string G\_CORE = "Absorption Settings"

abs\_volume\_lookback = input.int(20, "Volume Lookback Period", minval=10,  
group=G\_CORE)

abs\_zscore\_threshold = input.float(1.5, "Volume Z-Score Threshold", step=0.1, minval=1.0,

```
group=G_CORE)
abs_wick_threshold    = input.float(0.30,"Minimum Wick Size %",    step=0.05, minval=0.1,
maxval=0.9, group=G_CORE)
abs_delta_ltf        = input.timeframe("1", "Delta Timeframe",
group=G_CORE)
```

```
var string G_CONF = "Confidence Settings"
abs_min_confidence    = input.int(2,    "Minimum Stars to Display", minval=1, maxval=5,
group=G_CONF)
abs_require_multi_bar  = input.bool(true, "Require Multi-Bar Confirmation",
group=G_CONF)
abs_multi_bar_tolerance = input.float(0.3, "Multi-Bar Tolerance (% ATR)", step=0.1,
group=G_CONF)
```

```
var string G_VIS = "Visuals"
abs_show_bubbles      = input.bool(true, "Show Absorption Bubbles",
group=G_VIS)
abs_show_labels       = input.bool(true, "Show Confidence Labels",
group=G_VIS)
abs_show_dashboard    = input.bool(true, "Show Dashboard",
group=G_VIS)
abs_bubble_transparency = input.int(30,    "Bubble Transparency", minval=0, maxval=100,
group=G_VIS)
```

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```
// CALCULATIONS
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```
abs_atr = ta.atr(14)
```

```
[openVolume, maxVolume, minVolume, bar_delta] = ta.requestVolumeDelta(abs_delta_ltf)
```

```
avg_volume    = taxsma(volume, abs_volume_lookback)
stdev_volume  = taxstdev(volume, abs_volume_lookback)
```

```

volume_zscore = stdev_volume > 0 ? (volume - avg_volume) / stdev_volume : 0
volume_anomaly = volume_zscore >= abs_zscore_threshold

bar_range    = high - low
upper_wick    = high - math.max(open, close)
lower_wick    = math.min(open, close) - low
upper_wick_pct = bar_range > 0 ? upper_wick / bar_range : 0
lower_wick_pct = bar_range > 0 ? lower_wick / bar_range : 0

avg_delta    = ta.sma(bar_delta, abs_volume_lookback)
stdev_delta  = taxstdev(bar_delta, abs_volume_lookback)
delta_zscore = stdev_delta > 0 ? (bar_delta - avg_delta) / stdev_delta : 0

bullish_absorption = bar_delta < 0 and volume_anomaly and (close >= open)
bearish_absorption = bar_delta > 0 and volume_anomaly and (close <= open)

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// MULTI-BAR CONFIRMATION
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var float prev_bull_level    = na
var int  prev_bull_bar      = 0
var float prev_bear_level    = na
var int  prev_bear_bar      = 0
var bool multi_bar_bull_bonus = false
var bool multi_bar_bear_bonus = false

tolerance_range = abs_atr * abs_multi_bar_tolerance

if bullish_absorption
    current_level = low
    multi_bar_bull_bonus := not na(prev_bull_level) and math.abs(current_level - prev_bull_level) <=
tolerance_range and bar_index == prev_bull_bar + 1
    prev_bull_level := current_level

```

```
    prev_bull_bar := bar_index
else
    multi_bar_bull_bonus := false

if bearish_absorption
    current_level = high
    multi_bar_bear_bonus := not na(prev_bear_level) and math.abs(current_level - prev_bear_level)
    <= tolerance_range and bar_index == prev_bear_bar + 1
    prev_bear_level := current_level
    prev_bear_bar := bar_index
else
    multi_bar_bear_bonus := false
```

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```
// CONFIDENCE SCORING
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```
var float bull_confidence = 0.0
var float bear_confidence = 0.0
```

```
if bullish_absorption
    bull_confidence := 1.0
    if volume_zscore >= 2.0
        bull_confidence += 1.0
    if volume_zscore >= 3.0
        bull_confidence += 1.0
    if math.abs(delta_zscore) >= 2.0
        bull_confidence += 1.0
    if lower_wick_pct >= 0.4
        bull_confidence += 1.0
    if abs_require_multi_bar and multi_bar_bull_bonus
        bull_confidence += 1.0
    bull_confidence := math.min(bull_confidence, 5.0)
else
```

```

bull_confidence := 0.0

if bearish_absorption
  bear_confidence := 1.0
  if volume_zscore >= 2.0
    bear_confidence += 1.0
  if volume_zscore >= 3.0
    bear_confidence += 1.0
  if math.abs(delta_zscore) >= 2.0
    bear_confidence += 1.0
  if upper_wick_pct >= 0.4
    bear_confidence += 1.0
  if abs_require_multi_bar and multi_bar_bear_bonus
    bear_confidence += 1.0
  bear_confidence := math.min(bear_confidence, 5.0)
else
  bear_confidence := 0.0

bull_show = abs_show_bubbles and bullish_absorption and bull_confidence >=
abs_min_confidence
bear_show = abs_show_bubbles and bearish_absorption and bear_confidence >=
abs_min_confidence

//
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// VISUALIZATION
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get_bull_color(conf) =>
  color.from_gradient(conf, 1, 5, color.new(color.green, 70), color.new(color.green,
abs_bubble_transparency))

get_bear_color(conf) =>
  color.from_gradient(conf, 1, 5, color.new(color.red, 70), color.new(color.red,
abs_bubble_transparency))

```

```
bull_bubble_color = bull_show ? get_bull_color(bull_confidence) : na
bear_bubble_color = bear_show ? get_bear_color(bear_confidence) : na
```

```
plotshape(bull_show and bull_confidence >= 5 ? low : na, "Bull 5★", shape.circle, location.absolute,
bull_bubble_color, size=size×huge)
plotshape(bull_show and bull_confidence == 4 ? low : na, "Bull 4★", shape.circle, location.absolute,
bull_bubble_color, size=size×large)
plotshape(bull_show and bull_confidence == 3 ? low : na, "Bull 3★", shape.circle, location.absolute,
bull_bubble_color, size=size×normal)
plotshape(bull_show and bull_confidence == 2 ? low : na, "Bull 2★", shape.circle, location.absolute,
bull_bubble_color, size=size×small)
plotshape(bull_show and bull_confidence == 1 ? low : na, "Bull 1★", shape.circle, location.absolute,
bull_bubble_color, size=size×tiny)
```

```
plotshape(bear_show and bear_confidence >= 5 ? high : na, "Bear 5★", shape.circle,
location.absolute, bear_bubble_color, size=size×huge)
plotshape(bear_show and bear_confidence == 4 ? high : na, "Bear 4★", shape.circle,
location.absolute, bear_bubble_color, size=size×large)
plotshape(bear_show and bear_confidence == 3 ? high : na, "Bear 3★", shape.circle,
location.absolute, bear_bubble_color, size=size×normal)
plotshape(bear_show and bear_confidence == 2 ? high : na, "Bear 2★", shape.circle,
location.absolute, bear_bubble_color, size=size×small)
plotshape(bear_show and bear_confidence == 1 ? high : na, "Bear 1★", shape.circle,
location.absolute, bear_bubble_color, size=size×tiny)
```

```
if abs_show_labels and bull_show
    label_y    = low - (abs_atr * 0.5)
    label_text = "★ " + str.tostring(bull_confidence, "#") + " ★"
    tooltip_text = "BULLISH ABSORPTION\nConfidence: " + str.tostring(bull_confidence, "#") + "/"
5\nVolume Z: " + str.tostring(volume_zscore, "#.##") + "\nDelta Z: " + str.tostring(delta_zscore,
"#.##") + "\nLower Wick: " + str.tostring(lower_wick_pct * 100, "#.##") + "%\nNet Delta: " +
str.tostring(bar_delta, "#") + "\n" + (multi_bar_bull_bonus ? "✓ Multi-bar" : "")
    label.new(bar_index, label_y, label_text, color=color×new(color×green, 0), textcolor=color×white,
style=label.style_label_up, size=size×small, tooltip=tooltip_text)
```

```
if abs_show_labels and bear_show
    label_y    = high + (abs_atr * 0.5)
```

```
label_text = "★ " + str.toString(bear_confidence, "#") + " ★"  
tooltip_text = "BEARISH ABSORPTION\nConfidence: " + str.toString(bear_confidence, "#") + "/"  
5\nVolume Z: " + str.toString(volume_zscore, "###") + "\nDelta Z: " + str.toString(delta_zscore,  
"###") + "\nUpper Wick: " + str.toString(upper_wick_pct * 100, "#.#") + "%\nNet Delta: " +  
str.toString(bar_delta, "#") + "\n" + (multi_bar_bear_bonus ? "✓ Multi-bar" : "")  
label.new(bar_index, label_y, label_text, color=color.new(color.red, 0), textcolor=color.white,  
style=label.style_label_down, size=size.small, tooltip=tooltip_text)
```

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```
// DASHBOARD
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```
if abs_show_dashboard
```

```
var table dashboard = table.new(position.top_right, 2, 6, border_width=1,  
border_color=color.gray, frame_width=1, frame_color=color.gray)
```

```
table.cell(dashboard, 0, 0, "🎯 Absorption Monitor", bgcolor=color.new(color.blue, 70),  
text_color=color.white, text_size=size.normal)
```

```
table.cell(dashboard, 1, 0, "Value", bgcolor=color.new(color.blue, 70),  
text_color=color.white, text_size=size.normal)
```

```
table.cell(dashboard, 0, 1, "Volume Z-Score", bgcolor=color.new(color.black, 80),  
text_color=color.white, text_size=size.small)
```

```
table.cell(dashboard, 1, 1, str.toString(volume_zscore, "###"),  
bgcolor=color.new(volume_anomaly ? color.yellow : color.gray, 70), text_color=color.white,  
text_size=size.small)
```

```
table.cell(dashboard, 0, 2, "Delta Z-Score", bgcolor=color.new(color.black, 80),  
text_color=color.white, text_size=size.small)
```

```
table.cell(dashboard, 1, 2, str.toString(delta_zscore, "###"), bgcolor=color.new(delta_zscore >  
2 ? color.green : delta_zscore < -2 ? color.red : color.gray, 70), text_color=color.white,  
text_size=size.small)
```

```
table.cell(dashboard, 0, 3, "Net Delta", bgcolor=color.new(color.black, 80),
```

```
text_color=color×white, text_size=size×small)
```

```
table.cell(dashboard, 1, 3, str.tostring(bar_delta, "#"), bgcolor=color×new(bar_delta > 0 ?  
color.green : bar_delta < 0 ? color.red : color.gray, 70), text_color=color×white,  
text_size=size×small)
```

```
table.cell(dashboard, 0, 4, "Absorption", bgcolor=color.new(color.black, 80),  
text_color=color×white, text_size=size×small)
```

```
signal_text = bull_show ? "🟢 BULL " + str.tostring(bull_confidence, "#") + "★" : bear_show ?  
"🔴 BEAR " + str.tostring(bear_confidence, "#") + "★" : "---"
```

```
signal_color = bull_show ? color.green : bear_show ? color.red : color.gray  
table.cell(dashboard, 1, 4, signal_text, bgcolor=color.new(signal_color, 70),  
text_color=color.white, text_size=size×small)
```

```
table.cell(dashboard, 0, 5, "Multi-Bar Confirm", bgcolor=color.new(color.black, 80),  
text_color=color.white, text_size=size×small)
```

```
mb_active = multi_bar_bull_bonus or multi_bar_bear_bonus  
table.cell(dashboard, 1, 5, mb_active ? "✓ YES" : "---", bgcolor=color.new(mb_active ? color.teal :  
color.gray, 70), text_color=color.white, text_size=size.small)
```

```
bgcolor(volume_anomaly ? color.new(color.yellow, 97) : na, title="Volume Spike")
```